

Assignment 5: Data Visualization

Haochuan Zhan

Spring 2026

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1  
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
## Warning: package 'purrr' was built under R version 4.2.3

## Warning: package 'dplyr' was built under R version 4.2.3

## Warning: package 'stringr' was built under R version 4.2.3

## Warning: package 'forcats' was built under R version 4.2.3

## Warning: package 'lubridate' was built under R version 4.2.3

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at C:/Users/zhc/Desktop/R/EDE_Spring2026
```

```
library(cowplot)
```

```
## Warning: package 'cowplot' was built under R version 4.2.3
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
getwd()
```

```
## [1] "C:/Users/zhc/Desktop/R/EDE_Spring2026/Assignments"
```

```
ppl <- read_csv(
  here("Data", "Processed_KEY", "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  show_col_types = FALSE)
niwo <- read_csv(
  here("Data", "Processed_KEY", "NEON_NIWO_Litter_mass_trap_Processed.csv"),
  show_col_types = FALSE)
#2
class(ppl$sampleddate)
```

```
## [1] "Date"
```

```
class(niwo$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
my_theme <- theme_minimal() +
  theme(
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5)
  )
theme_set(my_theme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

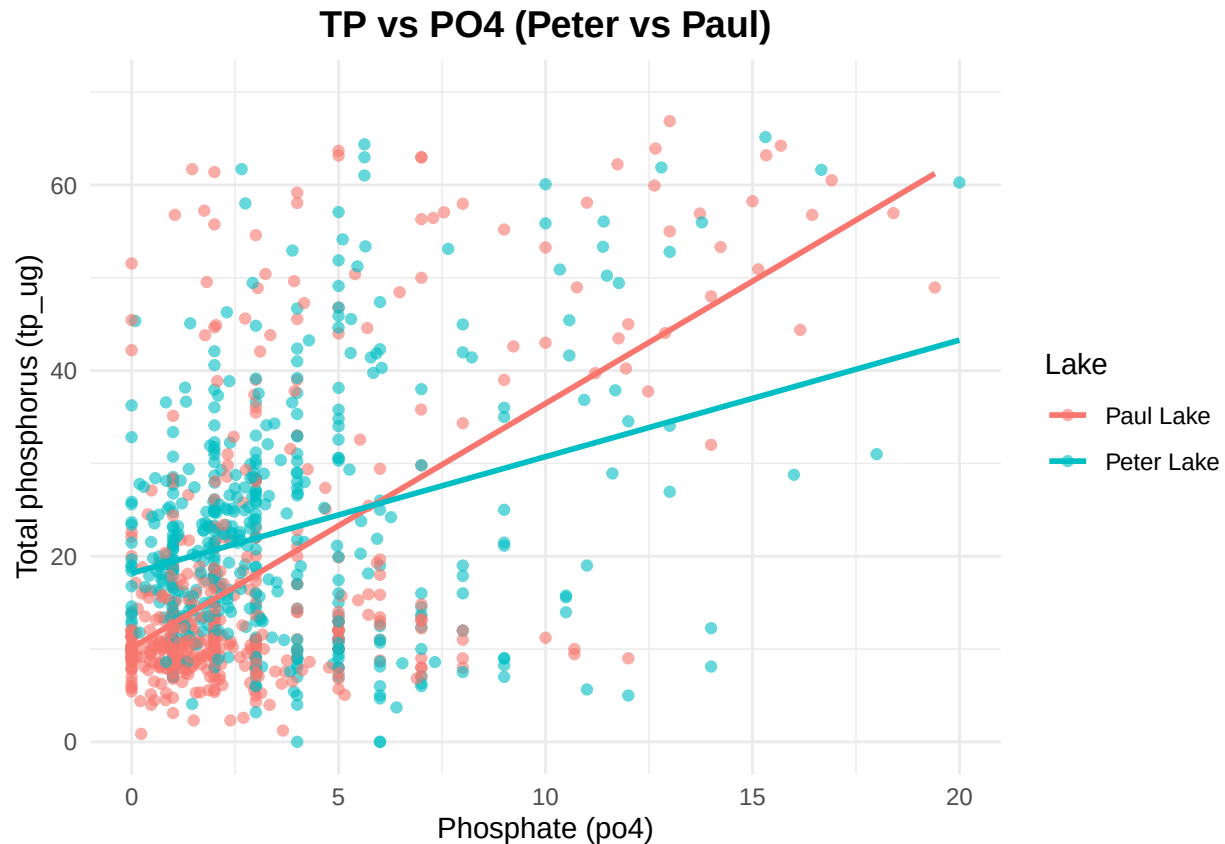
4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
ppl4 <- ppl %>% drop_na(po4, tp_ug, lakename)
ggplot(ppl4, aes(x = po4, y = tp_ug, color = lakename)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  xlim(0, 20) +
  ylim(0, 70) +
  labs(
    title = "TP vs P04 (Peter vs Paul)",
    x = "Phosphate (po4)",
    y = "Total phosphorus (tp_ug)",
    color = "Lake")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 61 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 61 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

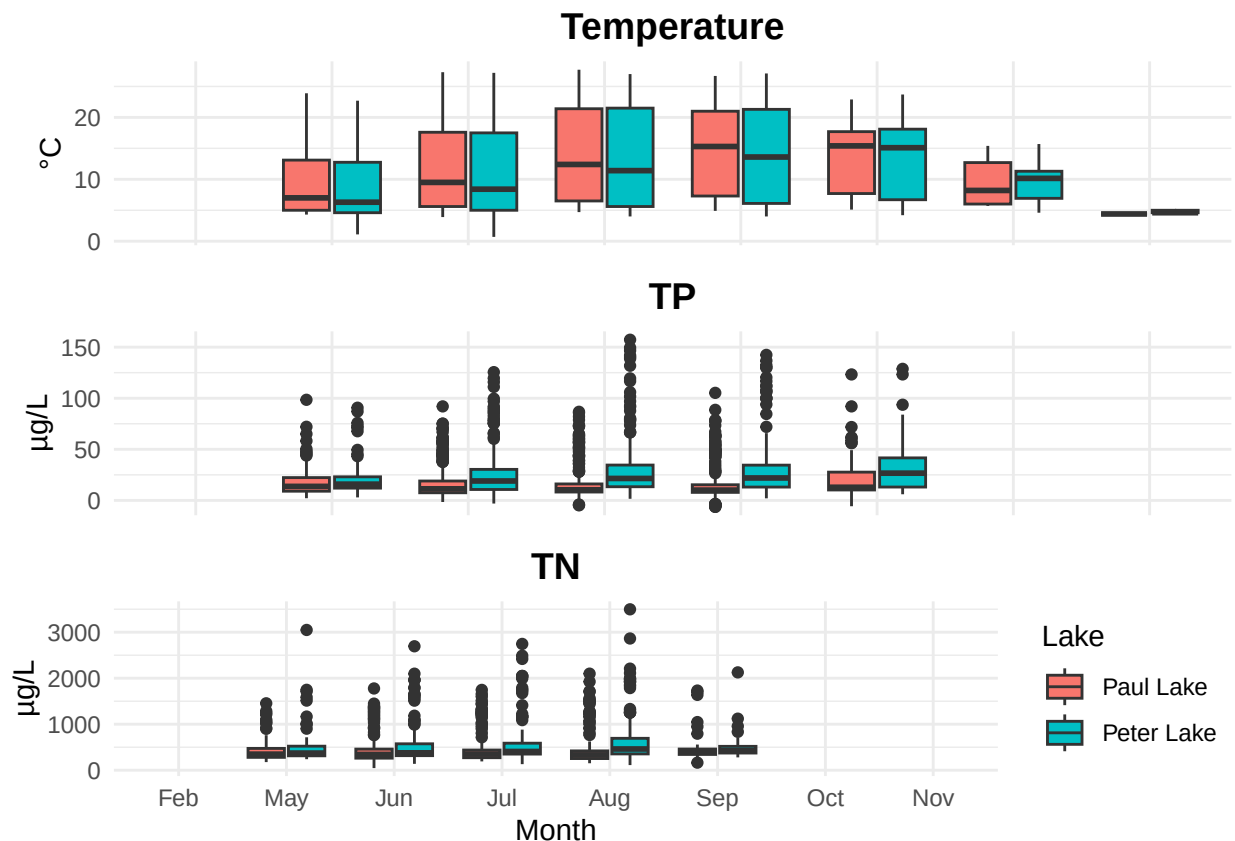
```
#5
library(cowplot)
ppl5 <- ppl %>%
  mutate(month = factor(month, levels = 1:12, labels = month.abb))
p_temp <- ggplot(ppl5, aes(x = month, y = temperature_C, fill = lakename)) +
  geom_boxplot() +
  labs(title = "Temperature", x = NULL, y = "°C", fill = "Lake") +
  theme(axis.text.x = element_blank(), legend.position = "none")
p_tp <- ggplot(ppl5, aes(x = month, y = tp_ug, fill = lakename)) +
  geom_boxplot() +
  labs(title = "TP", x = NULL, y = "µg/L", fill = "Lake") +
  theme(axis.text.x = element_blank(), legend.position = "none")
```

```
p_tn <- ggplot(pp15, aes(x = month, y = tn_ug, fill = lakename)) +
  geom_boxplot() +
  labs(title = "TN", x = "Month", y = "µg/L", fill = "Lake")
plot_grid(
  p_temp, p_tp, p_tn,
  ncol = 1,
  align = "v", axis = "l",
  rel_heights = c(1, 1, 1.2)
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



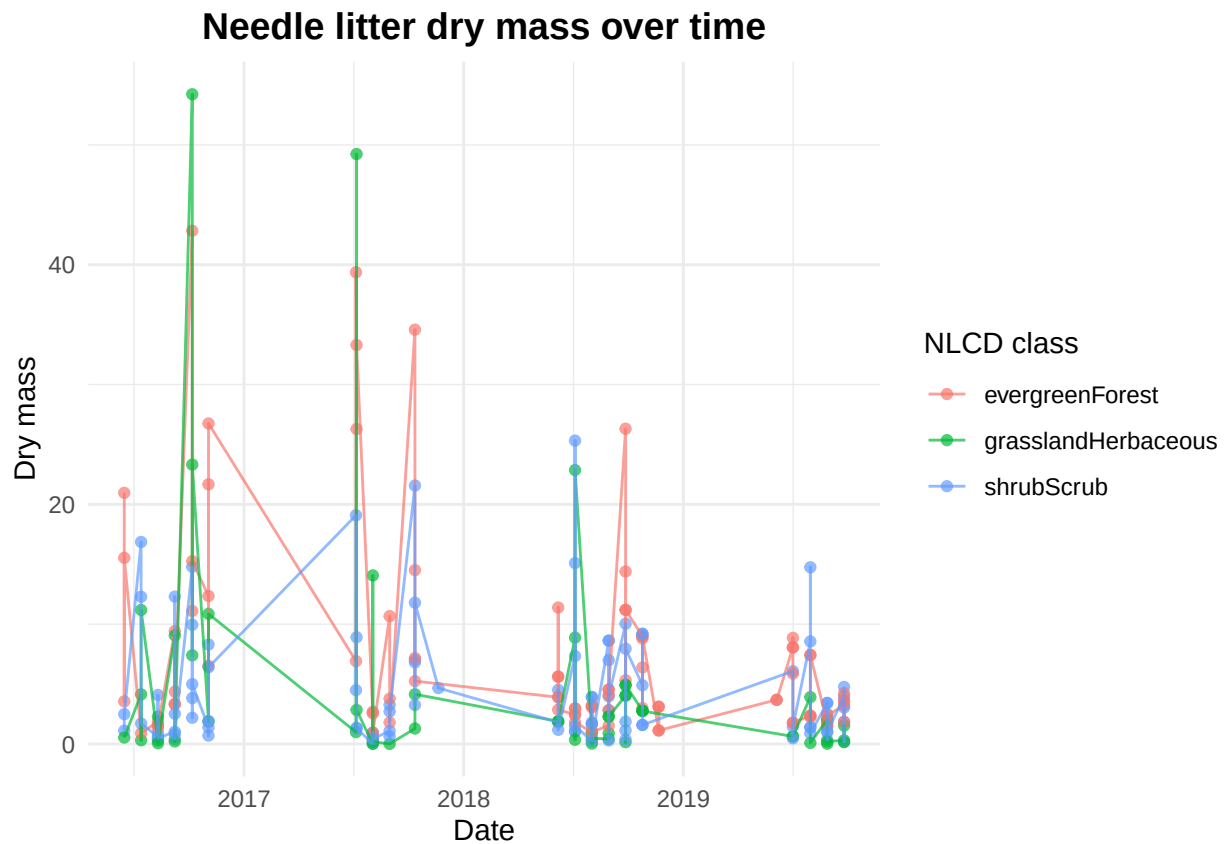
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Temperature is lowest in winter and highest in summer. TP and TN generally show higher values and more variability in the warm months than in cold months, and the two lakes differ, with one lake having consistently higher nutrient levels in several months.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

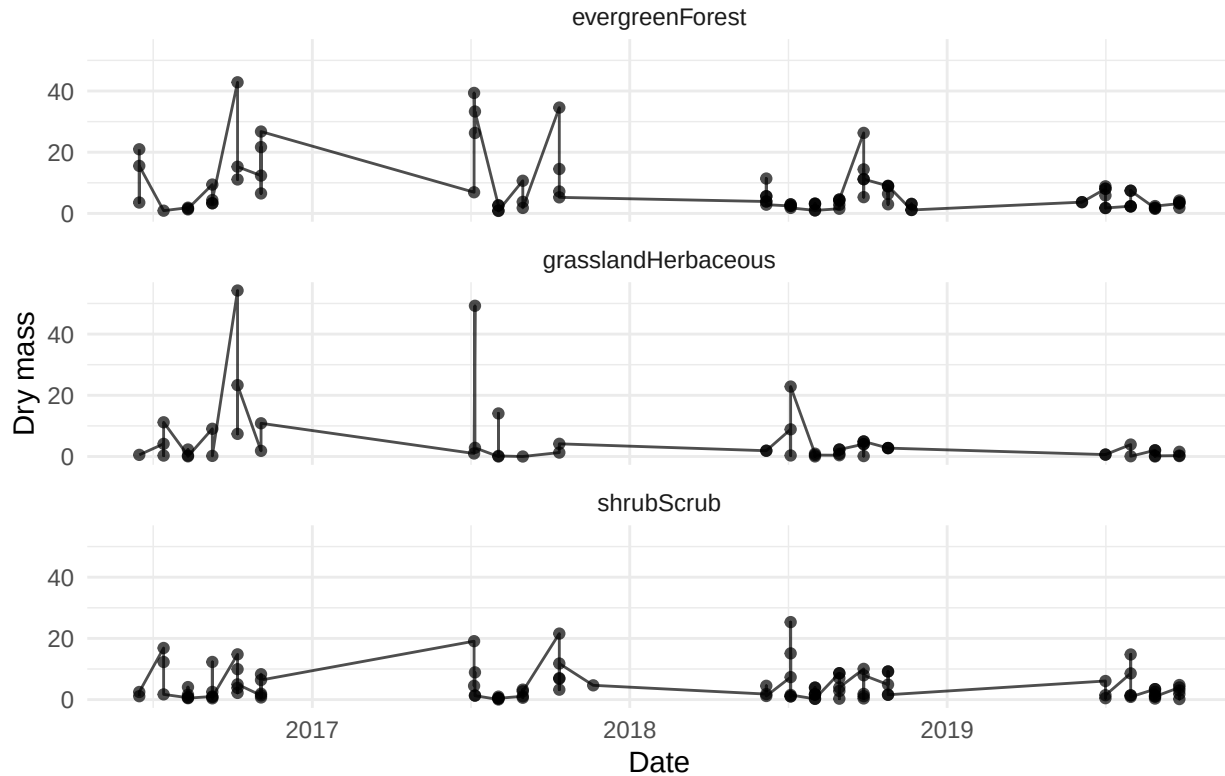
```
#6
niwo_needles <- niwo %>%
  filter(functionalGroup == "Needles")

ggplot(niwo_needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.7) +
  geom_line(alpha = 0.7) +
  labs(title = "Needle litter dry mass over time",
       x = "Date", y = "Dry mass", color = "NLCD class")
```



```
#7
ggplot(niwo_needles, aes(x = collectDate, y = dryMass)) +
  geom_point(alpha = 0.7) +
  geom_line(alpha = 0.7) +
  facet_wrap(~ nlcdClass, ncol = 1) +
  labs(title = "Needle litter dry mass over time (faceted by NLCD class)",
       x = "Date", y = "Dry mass")
```

Needle litter dry mass over time (faceted by NLCD class)



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think plot 7 is more effective because faceting separates the NLCD classes into their own panels, so trends and variability over time are easier to see without the overlap and visual clutter from multiple colored lines in one plot. Plot 6 is still useful for quick comparison across classes, but the overlapping points/lines make it harder to read.